ASSESSMENT A3 · STUDENT PACKAGE

Maths and Cognition Lab Report

Week 10 · Individual assessment

SUBMISSION	WEIGHT	COVERS	ASSIGNED CASE
Single PDF via LMS, end of Week 10	15% of Semester 1 grade	Weeks 7, 9 and 10	See your LMS profile

What this assessment is testing

A3 tests two things that no other assessment in this programme tests.

1. Can you measure AI behaviour yourself?

Part 1 asks you to run a controlled experiment: the same question, 20 times, at two different temperature settings, recording exactly what happens. You are measuring AI inconsistency directly, in your own domain, with your own data. When you build your capstone in Week 15, you will already have first-hand evidence of what changes when temperature changes.

2. Can you apply a judgement framework to a real failure?

Part 2 gives you a real AI failure case from your assigned domain. Real people were affected. Your job is to answer three specific questions: what did it cost, could it have been caught, and who was responsible? Getting comfortable naming these specifically, before you build your own system, is what separates an engineer who ships responsibly from one who does not.

Part 1: Temperature experiment

What is temperature?

Setting	What it does
Temperature 0	The model always gives the most likely answer. Ask the same question 10 times and you get the same response every time. Highly consistent, no variation.
Temperature 1	The model introduces randomness. The same question can produce different responses on different runs: sometimes correct but worded differently, sometimes taking a different angle entirely.

> **Why it matters for building AI systems**
>
> If you are building a system that must give consistent, reliable answers, such as a feedback assistant or a medical triage tool, you need to know whether temperature 0 is accurate enough to trust and whether temperature 1 introduces enough variation to cause problems. This experiment gives you that evidence first-hand.

How to run the experiment

Step	What to do
1	Open Google AI Studio (aistudio.google.com) and select Gemini 2.5 Flash. Turn off **Grounding with Google Search** in the right panel before you start.
2	Choose one yes/no question relevant to your assigned domain. It must have a clear, verifiable right answer, not a matter of opinion.
3	Set the temperature slider to 0. Type your question in the prompt box and click Run. Copy the response verbatim into your results table. Repeat 10 times without changing the prompt (runs 1–10).
4	Move the temperature slider to 1. Run the exact same prompt 10 more times (runs 11–20). Copy each response verbatim. Do not summarise or paraphrase.
5	Compute three statistics for each condition: **accuracy** (correct out of 10), **unique responses** (how many different answers appeared) and **mode** (the most common answer). Then write a conclusion of no more than 100 words stating the minimum accuracy threshold you would require before deploying in a real system, and why.

> **Important: Grounding with Google Search**
>
> Turn off Grounding with Google Search in the Google AI Studio right panel before running any prompt. If it is on, the model pulls results from the web and your responses will reflect web search, not temperature alone. The experiment will not be valid.

Part 2: Cognition case brief

Your assigned AI failure case is shown in your LMS profile. You do not choose the case; it has been assigned so that every student analyses a different situation. A one-page case brief is available on the LMS under **A3 Resources**. Read it before you write.

Section	What to cover
What the system did	What was the AI system, who built it, who deployed it, and what was it supposed to do?
What went wrong	What did the system actually do? Where was the human override point, and why was it missed?
Q1: Cost of being wrong	What was the real-world consequence when the AI made a mistake, and who was affected? Quantify it: numbers, scale or severity.
Q2: Could it be verified?	Was it possible for a human to check the AI's output before it was acted on? Was it actually checked? Answer both questions.
Q3: Who was accountable?	Name the specific role, organisation or institution accountable when it failed, not 'the AI company'. Explain why they were responsible.

Section	What to cover
Governance control	Name one governance framework, identify the relevant principle or category, and explain how it would have applied to this case.

How it is marked

Criterion	Marks	What we are looking for
Experiment: 20 runs, results table present	**20**	20 rows. Both temperature conditions labelled. All responses verbatim.
Statistical measures	**20**	Accuracy, unique responses and mode computed correctly for both conditions.
Written conclusion	**25**	Deployment conclusion drawn. Specific accuracy threshold stated as a number, not 'high accuracy'.
Judgement framework: Q1, Q2, Q3 with specificity	**25**	All three answered. Q1 quantified. Q3 names a specific role or institution. Answers reference the assigned case.
Governance: framework named and applied	**10**	Framework named, relevant principle or category identified, applied to this specific case.
Total	**100**	**Weighted to 15% of the semester grade.**

Common mistakes to avoid

- Fewer than 10 runs per condition, so the method does not hold up.
- No results table, only a written description of what happened.
- A conclusion that only describes ('temperature 0 was consistent, temperature 1 varied') without drawing a conclusion about deployment risk.
- No accuracy threshold stated as a number. 'High accuracy' is not specific enough.
- Writing about a different case from the one assigned.
- Q3 answered as 'the AI company'. Name which company, which role, and why they were responsible.
- A governance framework named but not connected to the case. Explain which principle applies and how.
- Grounding with Google Search left on in Google AI Studio.

Submission template

Complete both parts and export as a single PDF

Fill in Part 1 (results table, statistics and conclusion) and Part 2 (case brief) below. Export this document as a single PDF and submit via the LMS before your Week 10 session.

STUDENT NAME	STUDENT ID
BATCH / SECTION	ASSIGNED CASE ID
DOMAIN	DATE SUBMITTED

Part 1: Temperature experiment

Question used	
Copy it exactly	

Results table

Copy each response verbatim, exactly as it appears in Google AI Studio. Do not summarise or paraphrase.

Run	Temp	Response (copied exactly from Google AI Studio)	Correct? (Y/N)
1	**0**		
2	**0**		
3	**0**		
4	**0**		
5	**0**		
6	**0**		
7	**0**		
8	**0**		
9	**0**		
10	**0**		

Run	Temp	Response (copied exactly from Google AI Studio)	Correct? (Y/N)
11	1		
12	1		
13	1		
14	1		
15	1		
16	1		
17	1		
18	1		
19	1		
20	1		

Statistical summary

Measure	Temperature 0 (runs 1–10)	Temperature 1 (runs 11–20)
Accuracy (correct out of 10)		
Unique responses		
Mode (most common answer)		

Conclusion (maximum 100 words)

State the minimum accuracy threshold you would require before deploying this in a real system, as a number, and justify it.

Part 2: Cognition case brief

Answer each section below using the case brief assigned to you in your LMS profile. Do not write about a different case.

Assigned case	
Case ID and name	

What the system did

Name the AI system, who built it, who deployed it, and what it was supposed to do.

What went wrong

What did the system actually do? Where was the human override point, and why was it missed?

Q1: Cost of being wrong

Quantify the cost: numbers, scale or severity. Who was affected?

Q2: Could it be verified?

Was it possible for a human to check the AI's output before it was acted on? Was it actually checked? Answer both questions.

Q3: Who was accountable?

Name the specific role, organisation or institution accountable when it failed, and explain why. Do not answer 'the AI company'.

Governance control

Name the framework, identify the relevant principle or category, and explain how it applies to this case.

Pre-submission checklist

- [ ] My results table has 20 rows: 10 at temperature 0 and 10 at temperature 1.
- [ ] All responses in my results table are copied verbatim from Google AI Studio, not summarised.
- [ ] Both temperature conditions are clearly labelled in the results table.
- [ ] Grounding with Google Search was turned off before I ran any prompt.
- [ ] I have computed accuracy, unique responses and mode for both conditions.
- [ ] My conclusion states a specific accuracy threshold as a number, not ‘high accuracy’.
- [ ] My conclusion draws a judgement about deployment risk, not just a description of what happened.
- [ ] My Part 2 response is about the case assigned to me.
- [ ] Q1 quantifies the cost, and Q3 names a specific role, organisation or institution.
- [ ] My governance section names a framework and connects it to my specific case.
- [ ] Both parts are in a single PDF, not two separate files.